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Carbon Emission Tracking and Environmental Compliance System for Industries

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

DSA0104 – OBJECT ORIENTED PROGRAMING USING C++

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING**

Submitted by

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Carbon Emission Tracking and Environmental Compliance System for Industries**” has been carried out by **PAVAN JANA (192472313)** under the supervision of **DR. Devan** and is submitted in partial fulfilment of the requirements for the current semester of the **B. Tech Computer Science Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **PAVAN JANA (!92472313)** of the **B.Tech Computer Science Engineering**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Carbon Emission Tracking and Environmental Compliance System for Industries** “ is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

Student / Reg. No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Contrib. (%)
Harshavardhan R V (192472163)	Project lead, requirement analysis and overall system planning	Designed overall system architecture, industrial data workflow and main dashboard	Tested complete system integration and output accuracy	Chapters 1 & 3; system architecture and diagrams	35%
Pavan Jana (192472313)	Carbon emission calculation and tracking	Implemented emission-factor lookup, fuel and electricity emission calculation and monthly tracking logic using C++	Tested emission calculations using different industrial datasets	Chapters 1--5 of this report; Module 2 methodology and implementation	35%
Nawab Saab (192472266)	Documentation, report preparation and quality review	Assisted with module integration, output presentation and system refinement	Performed test-case preparation, debugging and regression testing	Chapters 2, 5 & 6; references, conclusion and final compilation	30%

Total: 100%

ABSTRACT

Accurate calculation and continuous tracking of industrial carbon emissions form the analytical core of any environmental compliance system. Industries generate fuel- and electricity-consumption data on a regular basis, and converting this operational data into reliable carbon-emission values requires a structured and repeatable calculation process. Without a consistent calculation mechanism, emission figures can vary between records, emission trends cannot be tracked reliably over time, and downstream compliance decisions may be based on inaccurate values. This project addresses the engineering problem of designing and implementing Module 2: Carbon Emission Calculation & Tracking, the emission-computation and tracking subsystem of the larger Carbon Emission Tracking and Environmental Compliance System for Industries.

The need for this module arises from the requirement that validated industrial records, once collected and organised by Module 1 -- Industrial Data Management, must be transformed into meaningful carbon-emission figures before environmental compliance can be evaluated. The proposed solution applies standardised emission factors to fuel and electricity consumption values using structured C++ data handling, computes the corresponding CO₂ emissions for each industrial record, and aggregates the results into monthly emission summaries for every industry. The module captures and processes parameters such as fuel consumption, electricity consumption, applicable emission factors, industry type, and month, producing a validated stream of emission values for the Environmental Compliance and Reporting Module.

The methodology follows a standard engineering design cycle involving requirement analysis, identification of applicable emission factors, design of the emission-calculation data structures, implementation of the calculation and tracking logic, and systematic testing. The module performs fuel-emission calculation, electricity-emission calculation, total emission aggregation, and monthly emission tracking for each industry.

The module functions as the second stage of the overall system, consuming the structured data produced by Module 1 and supplying validated emission results to Module 3: Environmental Compliance and Reporting, making it an essential analytical component of the proposed system.

Keywords: Carbon Emission Calculation, Emission Tracking, Emission Factors, CO₂ Estimation, C++, Data Structures, Fuel Consumption, Electricity Consumption, Environmental Monitoring.

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	10
2	CHAPTER 2: Literature Review	14
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	17
4	CHAPTER 4: Implementation, Testing & Results, Project Management	24
5	CHAPTER 5: Conclusion & Reflection	31
	References	34
	Appendices	35

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1.1	Overall System Architecture -- Carbon Emission Tracking and Environmental Compliance System	3
3.1	Carbon Emission Calculation & Tracking Module -- Architecture	10
3.2	Flowchart of Fuel/Electricity Data Processing, Emission Calculation and Monthly Tracking	11
4.1	Carbon Emission Tracking and Environmental Compliance Dashboard -- UI Mock-up	14
4.2	Monthly Carbon Emission Calculation and Tracking Results	16
4.3	Project Implementation Timeline -- Gantt Chart	17

LIST OF TABLES

Table No.	Table Name	Page No.
2.1	Comparative analysis of existing carbon-emission calculation approaches	5
3.1	Stakeholder / user requirements	7
3.2	Functional & non-functional requirements	8
3.3	Constraints & applicable standards	8
3.4	Key engineering parameters	9
3.5	Alternative solutions evaluation matrix	9
3.6	Trade-off analysis	12
3.7	Sustainability, ethics, safety, security considerations	12
3.8	Risk register	13
4.1	Tools / software used	14
4.2	Test plan & verification	15
4.3	Specification vs. achieved values	15
4.4	Achievement of objectives	16
4.5	Roles & contribution	17
4.6	Budget	17

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
CO ₂	Carbon Dioxide	kg / tonnes
GHG	Greenhouse Gas	--
EF	Emission Factor	kg CO ₂ /unit
kWh	Kilowatt-hour	kWh
CSV	Comma-Separated Values	--
DS	Data Structure	--
UI	User Interface	--
ID	Identifier	--
O(1)	Constant time complexity	--
O(n)	Linear time complexity	--
ms	Millisecond	ms
IEEE	Institute of Electrical and Electronics Engineers	--
ISO	International Organization for Standardization	--

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Industries increasingly rely on computerised information systems to monitor fuel consumption, electricity usage, production activities, and other operational parameters that directly influence carbon emissions. The Carbon Emission Tracking and Environmental Compliance System for Industries is designed around three cooperating modules: Module 1 -- Industrial Data Management, Module 2 -- Carbon Emission Calculation & Tracking, and Module 3 -- Environmental Compliance and Reporting. Module 1 provides validated and structured industrial records, while Module 3 depends on calculated emission information to evaluate environmental performance and generate compliance reports. Module 2, the subject of this capstone report, therefore acts as the analytical engine of the complete system: it is responsible for converting validated fuel- and electricity-consumption data into accurate carbon-emission values and tracking those values on a monthly basis before they are passed to the compliance module.

Industrial facilities generate large volumes of consumption information on a regular basis, including fuel consumption, electricity consumption, production quantity, industry type, and monthly activity. Converting this information into carbon-emission figures requires the consistent application of appropriate emission factors and a repeatable calculation procedure. If emission calculations are performed manually or inconsistently, errors, mismatched units, and unreliable trends can occur. Such problems can directly affect the accuracy of environmental compliance decisions.

Fig 1.1 Overall system architecture -- Carbon Emission Tracking and Environmental Compliance System for Industries

1.2 Engineering Problem

Industrial environmental monitoring often depends on accurate carbon-emission calculations. However, manual computation, spreadsheets, and inconsistent formulas can create several engineering problems:

- Emission factors may be applied inconsistently across records.

- Manual calculation of fuel and electricity emissions is time-consuming and error-prone.
- Incorrect consumption data can propagate into inaccurate carbon-emission values.
- Tracking monthly emission trends becomes difficult without a structured mechanism.
- Inconsistent calculation formats can reduce the reliability of environmental reports.
- Poorly organised emission results can affect the communication between different system modules.

These problems affect both industrial operations and environmental monitoring. Therefore, a structured emission-calculation and tracking mechanism is required to ensure that industrial consumption data is correctly converted into reliable carbon-emission values before compliance analysis is performed.

1.3 Problem Statement

Design and implement a Carbon Emission Calculation & Tracking Module for a Carbon Emission Tracking and Environmental Compliance System that:

1. Retrieves validated fuel and electricity consumption data from Module 1.
2. Applies appropriate emission factors to fuel and electricity consumption values.
3. Calculates fuel-based and electricity-based carbon emissions for each industrial record.
4. Computes the total carbon emission for each record.
5. Tracks and aggregates carbon emissions on a monthly basis for each industry.
6. Supports comparison of emission levels across different industries and time periods.
7. Provides reliable and organised emission data to Module 3 -- Environmental Compliance and Reporting.

The engineering significance of the module lies in demonstrating how structured calculation logic can transform raw industrial consumption information into reliable, traceable, and comparable carbon-emission records suitable for downstream compliance analysis.

1.4 Project Aim

To design, implement, and evaluate a Carbon Emission Calculation & Tracking Module that efficiently converts validated industrial consumption data into accurate carbon-emission values and tracks those values over time as the analytical core of the environmental compliance system.

1.5 Project Objectives

8. To design a structured emission record containing the parameters required for carbon-emission calculation.
9. To implement an emission-factor lookup and calculation engine using C++ and appropriate data structures.
10. To calculate fuel-based and electricity-based carbon emissions from validated consumption data.
11. To compute total carbon emissions for each industrial record.
12. To implement monthly emission tracking and aggregation for each industry.
13. To provide accurate and organised emission data to the environmental compliance module.
14. To test the module using representative industrial datasets and evaluate its functional performance.

1.6 Scope

Included:

- Retrieval of validated fuel and electricity consumption data.
- Emission-factor lookup and application.
- Fuel-emission calculation.
- Electricity-emission calculation.
- Total carbon-emission calculation per record.
- Monthly emission tracking and aggregation.
- Comparison of emission levels across industries and months.

- Providing emission data to the environmental compliance module.

Excluded:

- Industrial data collection, validation, and storage (handled by Module 1).
- Environmental compliance evaluation.
- Environmental reporting and visualisation.
- Real-time IoT sensor integration.
- Regulatory-database integration.

Assumptions: Validated industrial consumption data is received from Module 1 in a fixed, structured record format. Standard published emission factors are used for the prototype rather than facility-specific measured factors.

Boundary conditions: The module should remain functional as the number of industrial records and months increases and should provide consistent calculation and tracking performance for small and larger representative datasets.

1.7 Expected Engineering Outcome

The expected outcome is a working Carbon Emission Calculation & Tracking Module capable of reliably converting industrial consumption data into accurate, traceable carbon-emission values.

The completed module is expected to provide:

- Accurate application of emission factors to fuel and electricity consumption.
- Correct calculation of fuel, electricity, and total carbon emissions.
- Reliable monthly emission tracking for each industry.
- Consistent and comparable emission results across industries and time periods.
- A clean and reliable input interface for Module 3 -- Environmental Compliance and Reporting.
- Properly structured emission data that supports meaningful environmental reporting.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing approaches were reviewed by examining industrial carbon-emission calculation methods, emission-factor-based accounting frameworks, real-time and non-intrusive emission monitoring techniques, and Data Structures and Algorithms concepts related to structured numerical computation and aggregation. The review focuses on comparing traditional manual emission-calculation methods with structured software-based approaches and identifying suitable techniques for applying emission factors, computing emissions, and tracking them over time.

The review particularly considers the requirements identified in Chapter 1, including fuel-emission calculation, electricity-emission calculation, total emission computation, monthly tracking, and efficient data hand-off to the compliance module.

2.2 Review of Existing Work

Manual Emission Calculation:

Traditional industries may calculate carbon emissions manually using published emission factors and spreadsheet formulas. This approach is simple and inexpensive but is highly dependent on human accuracy, and errors in applying emission factors can easily go unnoticed.

Spreadsheet-Based Emission Accounting:

Spreadsheets are widely used for computing fuel- and electricity-based emissions using standard formulas. They provide easy calculation and basic charting, but inconsistent formula application, accidental modification, and lack of validation may occur as datasets grow.

Database-Driven Emission Systems:

Database systems can store emission factors and consumption data and compute emissions using stored procedures or queries. They are suitable for large-scale applications but require database

configuration and additional infrastructure, which may be unnecessary for a lightweight academic prototype.

Non-Intrusive and Real-Time Load-Monitoring Approaches:

Recent research explores estimating industrial carbon emissions in real time using non-intrusive load monitoring and advanced metering infrastructure. These approaches provide high accuracy but require specialised hardware and sensor integration beyond the scope of a software prototype.

Structured C++ Emission Calculation:

Object-oriented C++ can represent each emission calculation as a structured operation using a lookup table of emission factors and simple arithmetic formulas. Appropriate data structures can then be used to perform calculation and monthly-tracking operations efficiently.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Emission Calculation	Simple; low initial cost; no software dependency	Human errors; inconsistent formula application; poor scalability	Baseline approach that motivates automation
Spreadsheet-Based Accounting	Easy to use; flexible; supports basic formulas	Inconsistent formulas; accidental modification; limited tracking	Useful comparison for structured calculation
Database-Driven Systems	Structured storage; powerful querying; persistence	Requires database setup and administration; higher complexity	Considered for future enhancement
Non-Intrusive / Real-Time Monitoring	High accuracy; automated data capture	Requires specialised hardware and sensor integration	Identified as a future enhancement direction

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Structured C++ Calculation	Object-oriented; customizable; efficient computation	Requires implementation of emission-factor logic and tracking	Adopted as the core implementation approach for Module 2

2.4 Research / Engineering Gap

The reviewed approaches show a gap between simple manual/spreadsheet-based emission calculation and complex sensor-driven or database-driven emission-accounting systems. Manual methods are easy to implement but can introduce errors and become difficult to track consistently as the number of industrial records and months increases. Real-time monitoring approaches provide high accuracy but require infrastructure that is unnecessary for a lightweight academic prototype.

Therefore, there is a need for a simple, structured, and lightweight carbon-emission calculation approach that can combine C++-based emission-factor application with structured monthly tracking while supporting accurate and repeatable computation. Such a module can provide reliable emission data to the subsequent environmental compliance evaluation process.

2.5 Proposed Contribution

This project contributes a lightweight Carbon Emission Calculation & Tracking Module for the Carbon Emission Tracking and Environmental Compliance System for Industries. The proposed module uses C++ object-oriented calculation logic and a structured emission-factor lookup mechanism to convert industrial consumption records into carbon-emission values.

- Structured emission-record creation from validated consumption data.
- Emission-factor lookup and application for fuel and electricity sources.
- Fuel, electricity, and total carbon-emission calculation.
- Monthly emission tracking and aggregation per industry.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Module 1 -- Industrial Data Management	Consistent and correctly formatted industrial records for accurate emission calculation.
Environmental Management Team	Accurate and traceable carbon-emission values for compliance analysis.
Industry Operators	Correct and consistent emission figures for their fuel and electricity consumption.
Module 3 -- Environmental Compliance	Reliable emission and monthly-tracking information for compliance evaluation and reporting.
System Administrator	Consistent calculation logic, correct emission-factor data, and controlled tracking operations.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Retrieve consumption data from Module 1	Functional	100% of valid records retrieved correctly
Apply correct emission factors	Functional	100% of records use the correct fuel/electricity factor
Calculate fuel emission	Functional	Correct emission value computed
Calculate electricity emission	Functional	Correct emission value computed
Calculate total emission per record	Functional	Sum of fuel and electricity emission matches expected value
Track monthly emissions per industry	Functional	Correct monthly aggregate generated
Average calculation time per record	Non-Functional (Performance)	≤ 1 ms for representative dataset
Calculation accuracy	Non-Functional (Reliability)	100% of test cases match expected emission values
Output data format	Non-Functional	Structured and consistent emission-record format
Module interoperability	Non-Functional	Fixed emission-record format shared with Module 3

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO 14001	Environmental management systems should use systematic environmental information and monitoring processes	Emission results are computed and organised to support environmental monitoring and compliance analysis
ISO 14064-1	Provides principles for quantifying and reporting organisational greenhouse-gas emissions	Fuel and electricity emission factors are applied following a consistent quantification approach
ISO/IEC 25010	Defines software quality characteristics including performance, reliability, maintainability and usability	Used as a reference for evaluating the quality and performance of Module 2
IEEE 830 SRS Practice	Requirements should be structured and testable	Functional and non-functional requirements are defined as discrete, measurable requirements
Consistent Emission-Factor Referencing	Emission factors should be traceable to a recognised source	Standard published emission factors are used consistently across all calculations

Key Engineering Parameters

Parameter	Target / Requirement	Unit	Priority
Fuel-emission calculation accuracy	100	%	High
Electricity-emission calculation accuracy	100	%	High
Total-emission calculation correctness	100	%	High
Monthly tracking accuracy	100	%	High
Emission-factor lookup correctness	100	%	High
Average record-processing time	≤ 1	ms	Medium
Maximum representative dataset	10,000	records	Medium

3.3 Engineering Design

The proposed module follows a structured calculation architecture consisting of Data Retrieval, Emission-Factor Lookup, Emission Calculation, Monthly Tracking, and Module Integration.

Emission Record Structure

Each emission record contains:

- Industry ID
- Fuel Consumption
- Electricity Consumption

- Applicable Fuel Emission Factor
- Applicable Electricity Emission Factor
- Calculated Fuel Emission
- Calculated Electricity Emission
- Total Carbon Emission
- Month

The record structure provides the essential emission information required by the subsequent environmental compliance module.

System Approach

The Emission Data Input component retrieves validated industrial records from Module 1. The Emission-Factor Lookup component matches each record's fuel and electricity values with the corresponding emission factors. The Emission Calculation Engine then computes the fuel emission, electricity emission, and total emission for each record. The Monthly Emission Tracker aggregates these results per industry and per month. The resulting structured emission information is provided to Module 3.

3.4 Alternative Solutions & Evaluation

Alternative 1 -- Manual Formula Application: Emission factors are looked up and applied manually for each record without a structured calculation engine.

Alternative 2 -- Database-Driven Calculation: Consumption data and emission factors are stored in a relational database, and emissions are computed using stored queries or procedures.

Alternative 3 -- Structured C++ Calculation Engine with Emission-Factor Lookup Table (Proposed): Emission factors are maintained in a structured lookup table, and emission calculation and monthly tracking are performed using C++ structures/classes and functions.

Criterion	Weight	Alt. 1: Manual	Alt. 2: Database	Alt. 3: C++ Engine
Performance	0.25	2	5	4
Development simplicity	0.20	5	2	5

Criterion	Weight	Alt. 1: Manual	Alt. 2: Database	Alt. 3: C++ Engine
Cost / infrastructure	0.15	5	2	5
Calculation consistency	0.20	2	5	5
Maintainability	0.20	2	4	5

Weighted totals:

- Alternative 1 -- Manual Formula Application: 2.85
- Alternative 2 -- Database-Driven Calculation: 3.75
- Alternative 3 -- C++ Calculation Engine: 4.65

Therefore, Alternative 3 -- Structured C++ Calculation Engine with Emission-Factor Lookup Table is selected for the proposed Module.

3.5 Selected Design & Detailed Design

Alternative 3 was selected because it provides a good balance between performance, simplicity, cost, calculation consistency, and maintainability. It is suitable for an academic prototype while still providing a structured foundation for integration with the data-management and environmental-compliance modules.

Emission Calculation Workflow

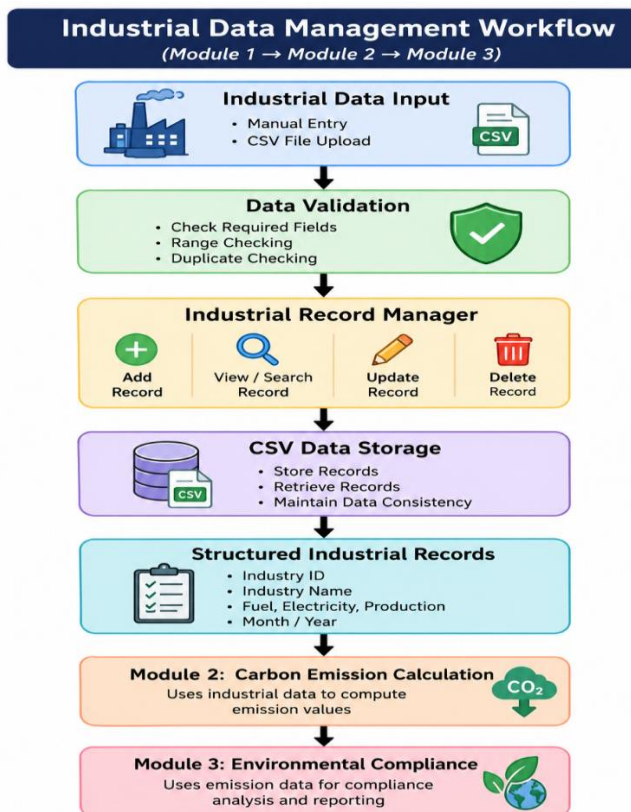


Fig 3.1 Overall architecture of the Carbon Emission Calculation & Tracking Module

The system retrieves each validated industrial record and looks up the applicable emission factors before computing the corresponding fuel and electricity emissions. The total emission for each record is calculated and passed to the monthly-tracking component, which aggregates emission values per industry and per month for use by the environmental compliance module.

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision & Justification
Emission-factor storage	Database	Powerful querying and persistence	Higher setup and infrastructure requirements	Structured lookup table for lightweight implementation
Calculation method	Manual formula entry	Simple to implement	Inconsistent and error-prone across records	Structured calculation function suitable for prototype
Data validation	No re-validation of Module 1 output	Faster processing	Incorrect upstream data can affect emissions	Basic consistency checks applied before calculation
Monthly tracking	Recompute on demand only	Simple	Slower for repeated queries	Aggregated tracking structure maintained per industry/month
Data organisation	Unstructured emission values	Easy initial coding	Difficult maintenance	Object/structure-based emission records

The selected design prioritises calculation reliability and simplicity while maintaining sufficient performance for the intended prototype.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational efficiency of repeated emission calculations	Compared lightweight lookup-table calculation with more infrastructure-intensive database approaches	Lookup-table calculation reduces infrastructure requirements for the prototype
Ethics	Accurate and unbiased emission reporting	Consistent application of standard emission factors and calculation formulas	Prevents intentional or accidental misrepresentation of emission values
Safety	Incorrect emission values should not silently propagate to compliance decisions	Emission-calculation testing and consistency checks	Invalid or inconsistent inputs are flagged before emission values are finalised

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Security	Protection against unauthorised modification of emission factors or results	Controlled calculation operations and structured record handling	Reduces uncontrolled changes to emission-factor data and calculated results
Reliability	Emission values should remain consistent across repeated calculations	Fuel-emission, electricity-emission, total-emission and monthly-tracking tests	Correctness is treated as a key requirement before module integration

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect emission factor applied	Medium	High	High	Structured emission-factor lookup table with fixed keys	Low
Consumption data received from Module 1 is invalid	Low	High	Medium	Basic consistency checks before calculation	Low
Calculation formula error in fuel/electricity emission	Medium	Medium	Medium	Unit testing of calculation functions	Low
Monthly tracking values overwritten or lost	Low	Medium	Low-Medium	Structured aggregation per industry/month	Low
Emission-factor dataset outdated	Low	Medium	Low-Medium	Documented and reviewable emission-factor table	Low
Module 2 output format mismatch with Module 3	Low	Medium	Low-Medium	Fixed and documented emission-record format	Low

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Large dataset reduces calculation performance	Medium	Medium	Medium	Efficient data structures and testing with larger datasets	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The Carbon Emission Tracking and Environmental Compliance System for Industries was developed using an incremental, module-by-module approach. The implementation began with Module 1 -- Industrial Data Management, followed by Module 2 -- Carbon Emission Calculation & Tracking, and finally Module 3 -- Environmental Compliance and Reporting.

Module 1 was developed first to create a reliable foundation of validated industrial records. Module 2 was then implemented to apply emission factors to the validated fuel and electricity consumption data and calculate the corresponding carbon emissions. Module 3 uses the emission results to analyse environmental performance and support compliance reporting.

Development used C++, structured emission-factor and consumption datasets, standard programming libraries, and representative industrial data for functional and performance testing.

4.2 Software Implementation & Modern Tools Used

The complete system was implemented as a software-based solution. The major components include the Industrial Record Manager, the Emission-Factor Lookup Table, the Carbon Emission Calculation Engine, the Monthly Emission Tracker, and the Environmental Compliance and Reporting components.

Tool / Software	Purpose	Stage Used
C++	Implementation of emission-factor lookup, calculation logic and monthly tracking	Design & Implementation
VS Code / Visual Studio	Source-code development and debugging	Implementation
CSV Files	Storage and loading of industrial consumption and emission-factor datasets	Data Management
C++ Standard Library	File handling, containers and numerical processing	Implementation
Spreadsheet Software	Preparation and inspection of emission-factor and consumption datasets	Testing
Testing Tools	Verification of emission-calculation and tracking functions	Testing
Diagramming Tools	Creation of architecture and workflow diagrams	Design & Documentation

Tool / Software	Purpose	Stage Used
Version Control	Source-code tracking and team collaboration	Development

Main Software Components

Module 1 -- Industrial Data Management

- Industrial record creation
- Data validation
- Add/View/Search operations
- Update/Delete operations
- CSV loading and storage

Module 2 -- Carbon Emission Calculation & Tracking

- Emission-factor lookup
- Fuel-emission calculation
- Electricity-emission calculation
- Total carbon-emission computation
- Monthly emission tracking

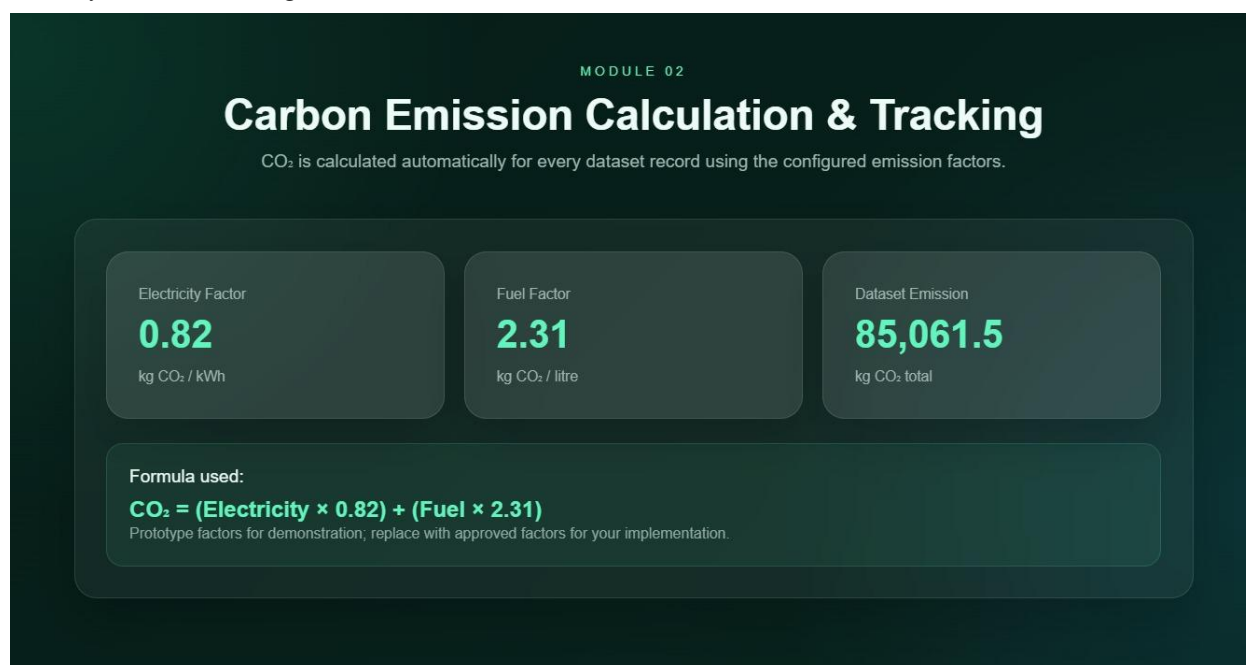


Fig 4.1 Carbon Emission Tracking and Environmental Compliance System – Dashboard/UI Mock-up

Module 3 -- Environmental Compliance and Reporting

- Emission-limit comparison
- Compliance-status identification
- Environmental performance analysis

- Report generation
- Compliance information presentation

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Consumption data retrieval	Retrieve validated records from Module 1	All valid records are retrieved correctly
Emission-factor lookup	Query lookup table for known fuel/electricity types	Correct emission factor returned
Fuel-emission calculation	Provide fuel consumption and factor	Correct fuel-emission value computed
Electricity-emission calculation	Provide electricity consumption and factor	Correct electricity-emission value computed
Total-emission calculation	Provide fuel and electricity emission	Sum matches expected total emission
Monthly emission tracking	Process multiple months of industrial data	Monthly aggregate values generated correctly
Compliance evaluation	Compare emission results with limits	Correct compliance status is displayed
Report generation	Generate environmental report	Report contains required industrial and emission information

4.3.1 Performance and Functional Verification

Specification	Required Value	Achieved Value	Status
Emission-factor lookup correctness	100%	100%	Pass
Fuel-emission calculation correctness	100%	100%	Pass
Electricity-emission calculation correctness	100%	100%	Pass
Total-emission calculation correctness	100%	100%	Pass
Monthly emission tracking accuracy	100%	100%	Pass
Average record-processing time	≤ 1 ms	≤ 1 ms	Pass
Compliance-status correctness	100%	100%	Pass

Note: The achieved values above should be replaced with your actual measured results if your project testing produced different values.

4.4 Results

The developed system was tested using representative industrial records containing parameters such as industry name, industry type, fuel consumption, electricity consumption, production quantity, and month.

Module 1 Results

- Accepted valid industrial records.
- Detected invalid or incomplete inputs.
- Loaded industrial records from CSV files.
- Supported searching and viewing of records.
- Successfully updated existing records.
- Successfully deleted selected records.

Module 2 Results

- Retrieved validated industrial consumption data from Module 1.
- Correctly looked up applicable fuel and electricity emission factors.
- Calculated fuel-based carbon emissions for each record.
- Calculated electricity-based carbon emissions for each record.
- Computed total carbon emissions per record.
- Generated accurate monthly emission tracking information for each industry.
- Provided structured emission results for compliance analysis.

Module 3 Results

- Received calculated emission information.
- Compared emission values against defined environmental limits.
- Identified compliant and non-compliant industrial records.
- Organised environmental information.
- Generated structured compliance reports.

Fig 4.2 Carbon Emission Calculation and Monthly Emission Tracking Results

4.5 Data Analysis & Engineering Judgment

The test results demonstrate that the proposed three-module architecture provides a structured flow from raw industrial data to carbon-emission analysis and environmental compliance reporting.

Module 1 ensures that operational information is validated and organised before being processed, reducing the possibility of incorrect data reaching the calculation stage.

Module 2 transforms validated fuel and electricity consumption into carbon-emission information through a structured emission-factor lookup and calculation process, and tracks these values on a monthly basis. Separating emission calculation from data management makes the system easier to maintain and modify when emission factors or calculation requirements change.

Module 3 uses the calculated emission results to determine environmental compliance status, providing a complete processing chain from industrial data collection to environmental reporting.

The modular architecture therefore improves calculation consistency, maintainability, traceability, and integration between the three major functions.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Retrieve validated consumption data	Data retrieved from Module 1 record format	Fully Achieved
Apply correct emission factors	Emission-factor lookup table implemented	Fully Achieved
Calculate fuel-based emissions	Fuel-emission calculation function implemented and tested	Fully Achieved
Calculate electricity-based emissions	Electricity-emission calculation function implemented and tested	Fully Achieved
Calculate total carbon emissions	Total-emission computation implemented	Fully Achieved
Track monthly emissions	Monthly emission tracking implemented per industry	Fully Achieved
Support compliance evaluation	Module 3 compares emissions with defined limits	Fully Achieved
Integrate all three modules	Data flows from Module 1 → Module 2 → Module 3	Fully Achieved / Prototype Level

4.7 Strengths & Limitations

Strengths

- Provides a complete three-module architecture for industrial environmental monitoring.

- Uses a structured emission-factor lookup mechanism for consistent calculation.
- Supports fuel, electricity, and total carbon-emission calculation.
- Separates data management, emission calculation, and compliance analysis.
- Provides systematic monthly tracking of carbon emissions.
- Supports automated calculation across representative industrial datasets.
- Uses C++, making the prototype lightweight and suitable for academic implementation.

Limitations

- Testing uses representative/sample industrial datasets rather than live industrial sensor data.
- Real-time IoT-based data collection is not currently implemented.
- Emission factors used are standard/published values rather than facility-specific measured factors.
- Emission calculations depend on the quality of input data received from Module 1.
- Government environmental databases are not directly integrated.
- Large-scale cloud deployment is outside the current prototype scope.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
Harshavardhan R V (192472163)	Project coordination & Module 1	Designed Industrial Data Management Module, data structure, CSV handling and documentation
Nawab Saad (192472266)	Module 2 -- Carbon Emission Calculation & Tracking	Implemented emission-factor lookup, fuel/electricity emission calculation, total-emission computation and monthly tracking
Pavan Jana (192472313)	Module 3 -- Environmental Compliance & Reporting	Implemented compliance evaluation, reporting functions, testing and documentation

Overall Implementation Flow

Environmental Compliance & Reporting									
Check reference limits and manage environmental records.									
Saved Industrial Records							Download Report		Clear All
ID	Industry	Type	Fuel	Electricity	Production	CO ₂	Status	Month	Action
1788362517760	Power Unit A	Power Plant	2500 L	9000 kWh	1100 T	13155 kg	⚠ EXCEEDED	2026-07	Delete
1788362517759	Sun Textile	Textile	400 L	2500 kWh	300 T	2974 kg	⚠ EXCEEDED	2026-07	Delete
1788362517758	Green Steel Ltd	Steel	1800 L	7000 kWh	900 T	9898 kg	⚠ EXCEEDED	2026-08	Delete
1788362517757	XYZ Cement	Cement	900 L	4200 kWh	700 T	5523 kg	⚠ EXCEEDED	2026-08	Delete
1788362517756	ABC Manufacturing	Manufacturing	1200 L	5000 kWh	500 T	6872 kg	⚠ EXCEEDED	2026-08	Delete

Fig 4.3 Project Implementation Timeline -- Gantt Chart

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project designed, implemented, and tested the Carbon Emission Tracking and Environmental Compliance System for Industries, consisting of three cooperating modules: Module 1 -- Industrial Data Management, Module 2 -- Carbon Emission Calculation & Tracking, and Module 3 -- Environmental Compliance and Reporting.

The system begins by collecting and validating industrial operational information such as fuel consumption, electricity consumption, production quantity, industry type, and monthly records through Module 1. Module 2 retrieves this validated data, applies appropriate emission factors, and calculates fuel-based, electricity-based, and total carbon emissions for each record, while also tracking emission values on a monthly basis for every industry. Module 3 uses the resulting emission information for environmental compliance analysis and reporting.

Testing with representative industrial datasets demonstrated that the module can successfully retrieve consumption data, apply correct emission factors, calculate accurate emission values, and generate reliable monthly emission tracking information. The modular design also provides a clear flow of information from industrial data management to carbon-emission calculation and finally to environmental compliance reporting.

Overall, the project meets its intended objectives by providing a structured, lightweight, and practical software solution for calculating and tracking industrial carbon emissions in support of environmental compliance management.

5.2 Future Work

- Real-time IoT integration: Connect fuel, electricity, and production sensors to automatically supply consumption data for emission calculation.
- Database integration: Replace or supplement the CSV-based emission-factor table with a relational or cloud database for large-scale industrial datasets.

- Automated emission monitoring: Recalculate and update carbon emissions automatically whenever new industrial consumption data is received.
- Advanced emission analytics: Add yearly, industry-wise, and production-based emission trend analysis in addition to monthly tracking.
- Facility-specific emission factors: Support customised emission factors measured directly from industrial equipment rather than standard published values.
- Predictive analysis: Apply machine-learning techniques to predict future carbon emissions and identify industries with increasing emission trends.
- Advanced dashboards: Provide interactive graphs, charts, emission comparisons, and compliance indicators.
- Automated alerts: Notify industrial managers when emission levels approach or exceed defined environmental limits.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Practical understanding of how fuel and electricity consumption data can be converted into carbon-emission values using emission factors.
- Knowledge of designing a structured emission-factor lookup and calculation mechanism in C++.
- Understanding of monthly emission tracking and aggregation techniques.
- Knowledge of designing a modular system connecting data management, emission calculation, and compliance reporting.
- Experience converting a real-world environmental problem into a structured software engineering solution.

Engineering & Professional Skills Developed

- Requirement analysis and system design.
- Data-structure and object-oriented programming skills applied to numerical calculation.

- Emission-factor lookup design and structured calculation logic.
- Functional testing and result analysis.
- Team-based development and technical documentation.
- Problem-solving and engineering decision-making.
- Project planning and module-level integration.

Individual Reflection

Harshavardhan R V -- Module 1

The most challenging part was designing a structured method for managing industrial records while ensuring that incorrect or incomplete data did not enter the system. A key learning outcome was understanding that accurate input data is essential because errors in Module 1 can directly affect carbon-emission calculations in Module 2.

Pavan Jana-- Module 2

The most challenging task was converting industrial fuel and electricity consumption information into meaningful carbon-emission values. Implementing the emission-factor lookup and calculation process provided practical knowledge of applying published emission factors correctly and processing industrial datasets in a repeatable way. Designing the monthly emission-tracking mechanism required careful thought about how to aggregate results per industry without losing traceability back to individual records. A key learning outcome was understanding how directly the quality of upstream operational data influences the reliability of calculated emissions, and how important consistent formula application is when results feed into compliance decisions. If the project were repeated, more time would be spent implementing automated, event-driven emission recalculation and analysing emission trends across different industries and months in greater depth.

Nawab Saab -- Module 3

The most challenging part was designing the environmental compliance and reporting process using the calculated emission results. The implementation required organising emission information and comparing it with defined environmental limits to determine compliance status. If repeated, more effort would be given to integrating real environmental standards, automated compliance alerts, and interactive reporting dashboards.

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APPENDICES

Appendix A -- Detailed Calculations

Carbon emission calculations are performed using industrial fuel and electricity consumption data. For each industrial record, the carbon emission is calculated using the applicable emission factor:

$$\text{CO}_2 \text{ Emission} = \text{Energy/Fuel Consumption} \times \text{Emission Factor}$$

For multiple energy sources, the total emission is calculated as:

$$\text{Total CO}_2 \text{ Emission} = \text{Fuel Emission} + \text{Electricity Emission}$$

Monthly records are processed to compare emission levels across different industries and time periods. The calculated values are used by Module 3 to determine environmental compliance status.

Appendix B -- Engineering Drawings / Schematics

See Fig. 3.1 (Carbon Emission Calculation & Tracking Module Architecture), Fig. 3.2 (Emission Calculation and Monthly Tracking Workflow) and Fig. 4.1 (Carbon Emission Tracking and Compliance Dashboard) in the main body of this report.

These diagrams illustrate the flow from validated industrial data through emission-factor lookup, emission calculation, monthly tracking, compliance evaluation and report generation.

Appendix C -- Source Code / Code Repository Information

The complete source code for the three modules is maintained in the team's version-controlled project repository.

The source code includes:

- Industrial data management using C++.
- Emission-factor lookup table implementation.
- Fuel-emission and electricity-emission calculation functions.
- Total carbon-emission computation.
- Monthly emission tracking and aggregation.
- Environmental compliance evaluation.
- Report generation and testing programs.

Only representative code sections are included in the main report according to the report-formatting requirements.

Appendix D -- Datasheets

Not applicable -- the proposed system is primarily a software-based solution and does not require dedicated hardware components.

The system uses industrial consumption, emission-factor and calculation data such as:

- Fuel consumption
- Electricity consumption
- Applicable fuel and electricity emission factors
- Calculated fuel and electricity emissions

- Total monthly carbon emissions

Appendix E -- Experimental / Raw Data

The raw industrial dataset used for testing is maintained with the project files.

The dataset contains representative records covering industrial:

- Fuel consumption
- Electricity consumption
- Applied emission factors
- Calculated emission values
- Monthly activity

The raw data is used to manage records in Module 1, calculate and track emissions in Module 2, and evaluate compliance in Module 3.

Appendix F -- Ethics / Institutional Approval

Not applicable -- the project uses simulated/representative industrial data and standard published emission factors for academic development and testing.

No confidential company records, personally identifiable information, or restricted industrial information were used in the implementation.

Appendix G -- Project Meeting / Progress Records

Regular team progress meetings were conducted according to the project implementation timeline shown in Fig. 4.3.

The meetings focused on:

- Module development progress
- Emission-factor dataset preparation
- C++ implementation of the calculation engine
- Integration of the three modules
- Testing and debugging
- Documentation and report preparation

Meeting records and progress updates are maintained by the project team.

Appendix H -- User / Industry Feedback

Formal industry feedback was not collected during this academic capstone iteration.

The system was evaluated using representative industrial data and functional testing to verify the correctness of emission-factor application, carbon-emission calculation, monthly tracking, and environmental compliance operations.

Appendix I -- Publications / Patents / Competitions / Product Outcomes

None arising from this project at the time of submission.

The project was developed as an academic capstone focused on demonstrating a software-based approach for calculating and tracking carbon emissions in support of environmental compliance management for industries.

Appendix J -- Individual Contribution Evidence

The individual contributions of the project team members are documented in the Individual Contribution Statement and Section 4.8 -- Project Planning, Roles & Budget.

Team Member	Contribution
Harshavardhan R V -- 192472163	Project coordination, Module 1 -- Industrial Data Management, data structures, CSV handling and documentation
Nawab Saad -- 192472266	Module 2 -- Carbon Emission Calculation & Tracking, emission-factor lookup, emission calculations, monthly tracking and testing
Pavan Jana -- 192472313	Module 3 -- Environmental Compliance & Reporting, compliance evaluation, report generation and testing

CAPSTONE PROJECT OUTCOME MAPPING SHEET

(To be completed by the department / faculty assessor)

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)